SmartWealth Technical Report

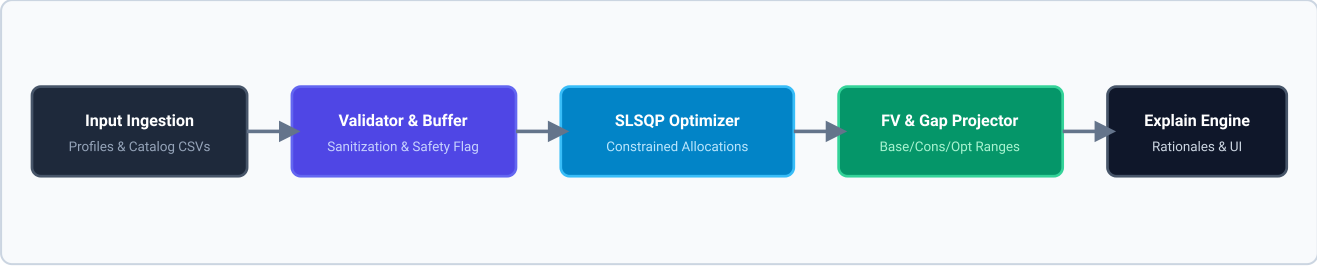
Goal-Based Wealth Engine & Asset Optimization System

Inter-IIT Tech Meet 15.0
NiveshNow Platform

Executive Summary

SmartWealth is an explainable, goal-based wealth planning engine built for NiveshNow. It transforms raw investor profiles and a multi-product catalog into mathematically defensible, risk-aware, and regulatory-compliant investment plans. Replacing black-box advice, SmartWealth integrates constrained convex optimization with automated horizon-based risk guardrails, cost transparency, and scenario-based feasibility modeling.

1. END-TO-END SYSTEM ARCHITECTURE



The system pipeline processes client inputs through five decoupled, functional modules designed for high reproducibility, deterministic execution, and complete auditability:

- **Ingestion & Schema Standardizer:** Normalizes heterogeneous input schemas, checks required data fields, enforces numerical bounds, and identifies profile inconsistencies.
- **Constraint-Bounded SLSQP Optimizer:** Formulates portfolio allocation as a constrained quadratic optimization problem, balancing return maximization against regulatory and liquidity limits.
- **Goal Feasibility & Gap Engine:** Evaluates discrete future value trajectories across three return bands and calculates exact monetary gap remedies.
- **Explainability Engine:** Automatically converts structural constraints and asset traits into clear, user-facing plain-language explanations.

2. DATA INGESTION & SAFETY GUARDRAILS

Before allocation, inputs undergo multi-pass validation. Profiling checks ensure that non-finite values (NaN, nulls) or negative goal amounts trigger early exits. Critical safety rules evaluate emergency liquidity: if reported emergency fund coverage is below 3.0 months, the system issues a prominent warning banner and prioritizes liquid cash reserves before committing capital to long-term investment products.

3. PORTFOLIO OPTIMIZATION ENGINE

SmartWealth frames portfolio construction as a **Constrained Convex Optimization** problem, solved using Sequential Least Squares Programming (SLSQP). This prevents arbitrary heuristic allocations while guaranteeing strict compliance with regulatory and investor-specific constraints.

Mathematical Formulation

Let $\mathbf{w} = [w_1, w_2, ..., w_N]^T$ represent the portfolio weight vector across N available products, where r_i is the expected return of product i, and R_i is its risk score.

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Maximize: r_p = ∑ (w_i × r_i) [for i = 1 to N]
Subject to:
1. ∑ w_i = 1.0 (Full Capital Allocation)
2. 0.0 ≤ w_i ≤ 0.40 (Single-Product Diversification Cap)
3. ∑ (w_i × R_i) ≤ Risk_Tolerance + 0.25 (Weighted Risk Limit)
4. w_i = 0 ∀ i where LockIn_months_i > Goal_Horizon_months
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4. REGULATORY & HORIZON-BASED GUARDRAILS

To prevent inappropriate equity exposure for short-term liabilities, the optimization engine enforces mandatory asset-class allocation bands based on the investor's goal horizon. When an investor's personal risk tolerance requires a stricter limit than the horizon rule, the stricter limit prevails.

Goal Horizon	High-Risk Cap (Risk 4–5)	Low-Risk Floor (Risk 1–2)	Regulatory Rationale
≤ 36 Months	Maximum 10%	Minimum 70%	Capital preservation; protects near-term capital from equity volatility.
37 – 84 Months	Maximum 30%	Minimum 45%	Balanced growth; provides stability with moderate inflation protection.
> 84 Months	Maximum 55%	Minimum 25%	Long-term wealth compounding; captures equity growth premium over cycles.

5. DETERMINISTIC CONFLICT RESOLUTION & FALLBACK

In edge cases where an investor's constraints are over-constrained (e.g., an ultra-short 6-month horizon paired with a very restrictive risk tolerance where no multi-asset combination satisfies all constraints under the 40% single-product cap), the SLSQP solver flags a constraint conflict. Rather than failing or violating safety boundaries, SmartWealth automatically triggers a **Safe Fallback Mechanism**: allocating 100% of capital to the lowest-risk, ultra-liquid product (Liquid Cash Reserve) and prompting user clarification.

6. GOAL FEASIBILITY & FUTURE VALUE METHODOLOGY

To provide realistic financial trajectories without falsely claiming guaranteed returns, SmartWealth models future wealth accumulation using discrete monthly compounding across three distinct return scenarios.

Future Value (FV) Compounding Formula

For initial investment **P**, monthly SIP **S**, annual return rate **r**, and horizon **n** months:

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FV(P, S, r, n) = P × (1 + r/12)^n + S × [ ((1 + r/12)^n - 1) / (r/12) ] (for r > 0)
FV(P, S, 0, n) = P + (S × n) (for r = 0)
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- **Base Scenario:** r_base = r_p (Blended Portfolio Return)
- **Conservative Scenario:** r_cons = max(r_p - 0.03, 0.01)
- **Optimistic Scenario:** r_opt = r_p + 0.03

7. GOAL GAP ANALYSIS & ACTIONABLE REMEDIATION

The goal gap is defined in nominal terms as **Goal Gap = Target Amount - Base FV**. When a shortfall is detected (Goal Gap > 0), the engine automatically calculates two concrete, mathematical remedies:

Remedy Option A

SIP Contribution Delta (ΔS)

Calculates the exact incremental monthly contribution required to close the gap over the original horizon.

Remedy Option B

Horizon Extension (Δn)

Solves iteratively for the additional months needed to reach the target amount under the existing monthly SIP rate.

8. COST STRUCTURE & LIQUIDITY ANALYSIS

SmartWealth enforces strict cost and liquidity transparency across all recommendations:

- **Weighted Total Expense Ratio (TER):** TER_p = ∑ (w_i × TER_i). The annual product cost in rupees is calculated as TER_p × Initial Capital.
- **Liquidity Coverage Verification:** The engine sums allocations in highly liquid assets (settlement ≤ 3 days) and compares them against the user's near-term liquidity need, flagging any shortfall.

9. EXPLAINABILITY ENGINE

To build investor trust, the explainability module converts mathematical outputs into transparent, compliance-friendly rationales. For every allocated asset, it details its functional role, its primary market downside, and explicit reasons why tempting higher-yielding alternatives were restricted or excluded by lock-in or horizon rules.

10. SYSTEM ASSUMPTIONS & ENGINEERING TRADE-OFFS

Building a robust production wealth engine requires explicit trade-offs between mathematical precision, computational efficiency, and user transparency.

Architecture Dimension	Design Choice Adopted	Engineering Trade-off & Rationale
Optimization Algorithm	SLSQP Convex Optimizer	Slightly higher compute overhead than rule-based heuristics, but guarantees globally optimal allocations under complex multi-constraint bounds.
Risk Management	Hard Constraint Bounds	Hard caps strictly enforce regulatory limits, sacrificing potential return upside to eliminate unacceptable downside risk for short horizons.
Return Modeling	Parametric Multi-Scenario Bands	Uses deterministic ±3% return bands rather than stochastic Monte Carlo to ensure fast, predictable execution suitable for interactive web applications.
Asset Allocation	Single-Product Cap (40%)	Prevents concentration risk in high-return assets, enforcing structural multi-asset diversification across all client portfolios.

11. EDGE CASES & FAILURE MODE RESOLUTION

The system was stress-tested against severe financial edge cases to ensure graceful degradation:

- **Inadequate Emergency Reserve (< 3 Months):** Prominently flagged in UI; system prompts user to fund a liquid reserve prior to long-term investing.
- **Lock-In Exceeding Horizon:** Hard-masked to 0% weight in optimizer bounds, eliminating liquidity lockup risks.
- **Market Shock Simulation (-15% Drawdown):** Tested via What-If module, recalculating updated goal gaps and SIP adjustments without breaking optimization limits.
- **Schema & Data Malformation:** Caught during ingestion by standardized mapping layers, returning explicit validation errors rather than corrupting calculations.

12. PRODUCTION ROADMAP & FUTURE ENHANCEMENTS

With core suitability and optimization established, future iterations will expand capabilities:

Phase 1: Advanced Modeling

Integrate stochastic Monte Carlo engine with sequence-of-returns risk modeling and historical stress testing.

Phase 2: Tax & Dynamic Rebalancing

Implement tax-aware asset location (STCG/LTCG optimization) and automated glide-path rebalancing as goal deadlines approach.

SmartWealth Technical Specification Document

Engineered for NiveshNow Digital Wealth Platform • Inter-IIT Tech Meet 15.0 Submission

All return projections are educational simulations based on supplied parameters and do not constitute guaranteed returns.